

Anatomy of AI Agent

Understanding the Building Blocks of Agentic AI Systems



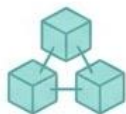
What We'll Dissect Today



1. Evolution from Chatbots to Agents



2. The Agentic Loop



3. Core Components



4. Complete Architecture



5. Takeaways & Next Steps

Why Agentic AI Matters

Chatbots



Reactive: Respond to specific prompts.



Shift from Reaction to Autonomy

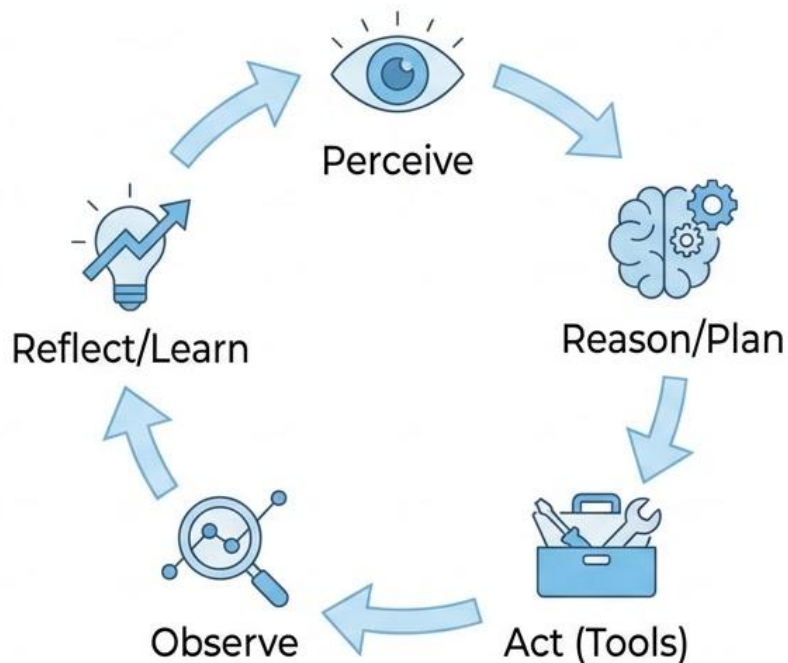
AI Agents



Autonomous: Act, plan, and use tools.

The Agentic Loop

The Heartbeat of Every AI Agent



The Brain: LLM as Reasoning Engine



- Core decision maker
- Understands goals & reasons
- Powers autonomous agents

Goals & Agent Profile



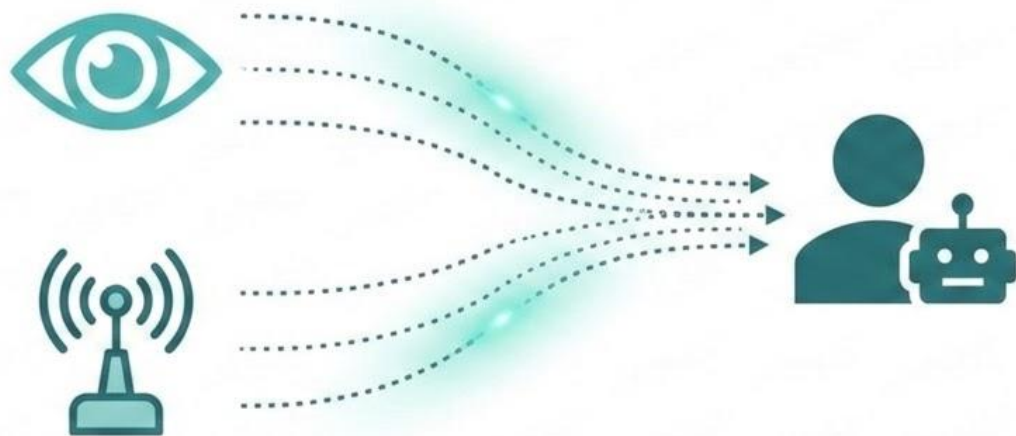
Goals & Definition

- Defines purpose, role & constraints



- Set via System Prompt
- Example: Test Automation Agent

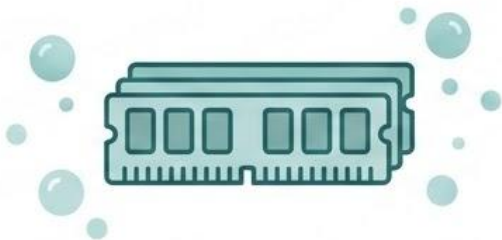
Perception: Sensing the Environment



- Queries, APIs, Documents
- NLP & Multimodal
- Connects world to reasoning

Memory: Short-Term & Long-Term

Short-Term



Short-Term

- Temporary storage for immediate tasks
- Limited capacity
- Data is quickly overwritten or cleared

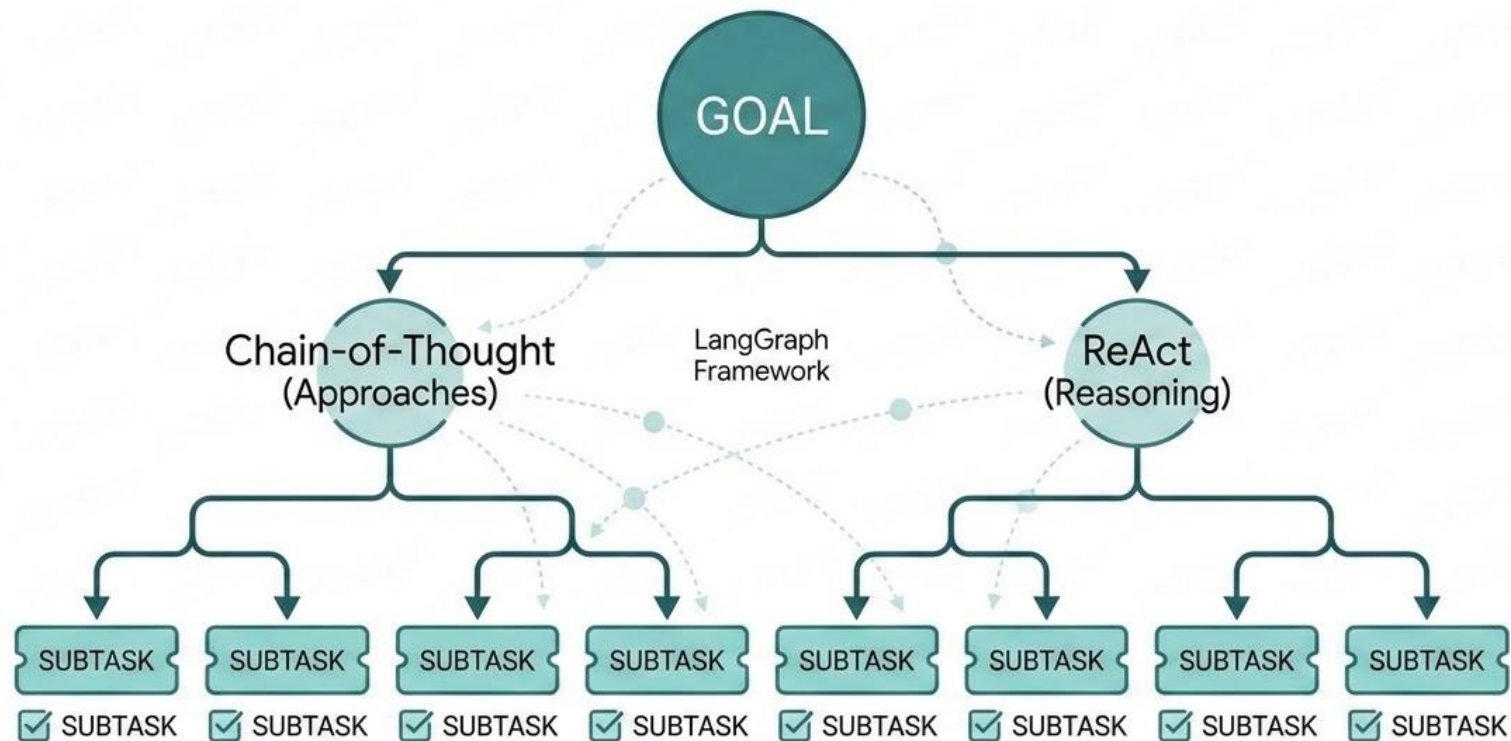
Long-Term



Long-Term

- Persistent, durable storage
- Large, theoretically unlimited capacity
- Data is indexed, retrieved, and reinforced

Planning: Breaking Down Tasks



Tools & Action: Interacting with the World



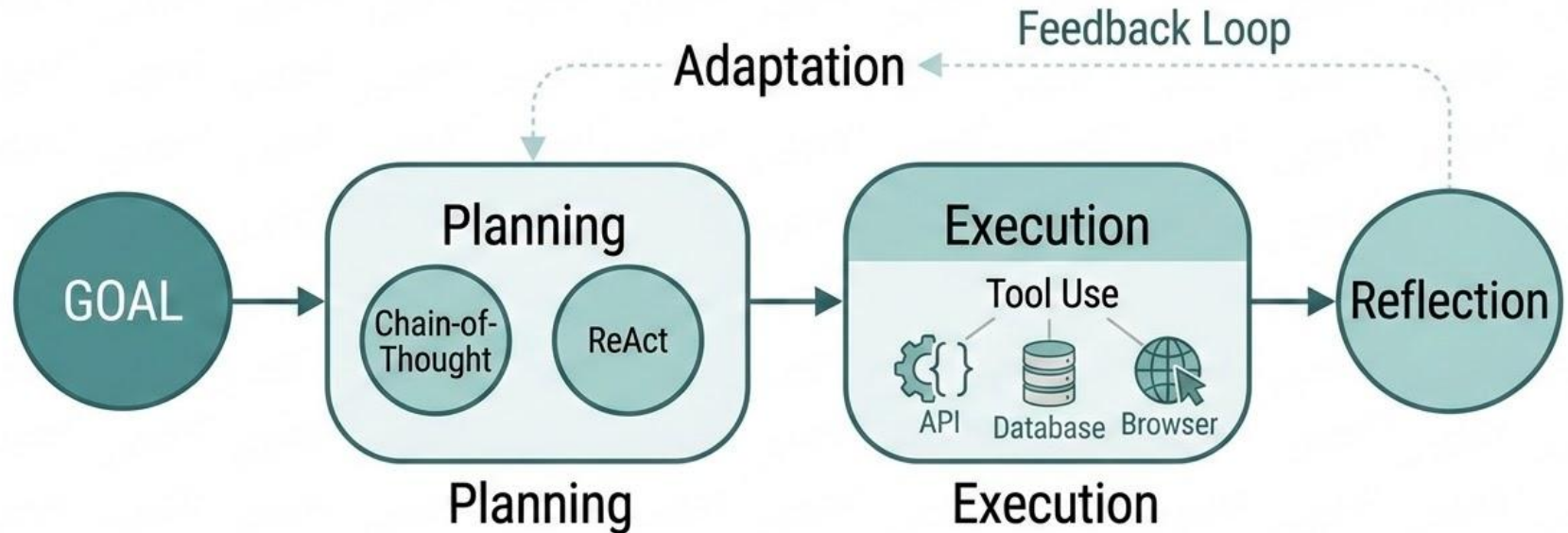
Tool Use (Execution)

- Tools are interfaces enabling interaction with external environments.
- Execution converts planning decisions into operational tasks.
- Allows information retrieval, state updates, and user interaction.

Tools (Examples)

- **API:** Connecting to external services and data.
- **Database:** Accessing and modifying structured data.
- **Browser:** Interacting with web pages and interfaces.

Reflection, Adaptation & Complete Anatomy



- Continuous loop for continuous learning.
- Adaptation optimizes future performance.
- Reflection analyzes results and plans.
- Adaptation optimizes future performance.

Key Takeaways & Next Steps

- ✓ **AGENT CORE (BRAIN):**
Decision-making logic and goal prioritization.
- ✓ **CONTEXT (MEMORY):**
Efficient storage and retrieval of relevant data.
- ✓ **STRATEGY (PLANNING):**
CoT and ReAct techniques for systematic action.
- ✓ **EXECUTION (TOOLS):**
Integrated API, Database, and Browser interfaces.
- ✓ **ITERATION (ADAPTATION):**
Feedback loops optimize future decisions.
- ✓ **IMPROVEMENT (REFLECTION):**
Systematic evaluation to refine entire system.

